

Application of transport modelling tools (ACP5-b) 2024/25: Temperature simulations in shallow ponds

Objectives:

The aim is to test if existing lake models can simulate water temperature in small and shallow water bodies (ponds). You will apply the LakeEnsemblR – a framework that facilitates running ensembles of five different vertical one-dimensional hydrodynamic lake models (Moore et al., 2021).

All models were developed and tested for larger lakes. Currently it is unclear to what extent they can predict water temperature in shallow ponds. Therefore, the model ensemble will be applied to a pond mesocosms from the EERES field station, for which temperature measurements are available. The simulations will cover a full seasonal cycle and will consider site specific forcing conditions (meteorological data). The model results will be compared to available measurements of water temperature.

The following sources are available:

R-Script and lake models	(Moore et al., 2021)
Characteristics of the pond mesocosms (including size/dimensions)	(Stehle et al., 2022)
Meteorological data	Measured at EERES (2020-2021)
Water temperature time series at different depths	(21.07.2020 – 21.06.2021, with some data gaps)
Characterization of temperature dynamics, thermal stratification and mixing:	(Holgerson et al., 2022; Read et al., 2011; Read et al., 2014)

You can work in groups focusing on the following group-specific objectives:

1. Model validation using observed water temperature: what systematic and random discrepancies exist between models and measurements?
2. Simulation of water temperature in scaled basins (original size of the mesocosms, x10, x100): What is the influence of basin size on water temperature and thermal stratification dynamics?
3. Analyze inter-annual variability (simulations with meteorological data from several years, or data from a weather generator): What is the influence inter-annual variations in meteorological conditions on the seasonal dynamics of temperature and thermal stratification in the ponds?
4. Simulation of climate change scenarios: How will water temperature and thermal stratification change in future climates?

Note: When answering the above-mentioned research questions, the focus should be on the methodological aspects of the respective model applications and the necessary analyses. A detailed answer to the questions in the context of other scientific work goes beyond the scope of this course.

Portfolio: Description of methods, results and a (short) discussion. The important thing is the complete documentation of the conducted model simulations.

Deadline for portfolio submission: **28.3.2025**

Meetings: on demand. Do you prefer fixed appointments?

Data und documents (including publications listed below):

<https://seafire.rlp.net/d/e9ec4043ca25495bb8d0/>

(you can also upload results to this folder for sharing)

References:

- Holgerson, M. A., Richardson, D. C., Roith, J., Bortolotti, L. E., Finlay, K., Hornbach, D. J., Gurung, K., Ness, A., Andersen, M. R., Bansal, S., Finlay, J. C., Cianci-Gaskill, J. A., Hahn, S., Janke, B. D., McDonald, C., Mesman, J. P., North, R. L., Roberts, C. O., Sweetman, J. N., & Webb, J. R. (2022). Classifying Mixing Regimes in Ponds and Shallow Lakes. *Water Resources Research*, 58(7), e2022WR032522. <https://doi.org/10.1029/2022WR032522>
- Moore, T. N., Mesman, J. P., Ladwig, R., Feldbauer, J., Olsson, F., Pilla, R. M., Shatwell, T., Venkiteswaran, J. J., Delany, A. D., Dugan, H., Rose, K. C., & Read, J. S. (2021). LakeEnsemblR: An R package that facilitates ensemble modelling of lakes. *Environmental Modelling & Software*, 143, 105101. <https://doi.org/10.1016/j.envsoft.2021.105101>
- Read, J. S., Hamilton, D. P., Jones, I. D., Muraoka, K., Winslow, L. A., Kroiss, R., Wu, C. H., & Gaiser, E. (2011). Derivation of lake mixing and stratification indices from high-resolution lake buoy data. *Environmental Modelling & Software*, 26(11), 1325-1336. <https://doi.org/10.1016/j.envsoft.2011.05.006>
- Read, J. S., Winslow, L. A., Hansen, G. J. A., Van Den Hoek, J., Hanson, P. C., Bruce, L. C., & Markfort, C. D. (2014). Simulating 2368 temperate lakes reveals weak coherence in stratification phenology. *Ecological Modelling*, 291, 142-150. <https://doi.org/10.1016/j.ecolmodel.2014.07.029>
- Stehle, S., Manfrin, A., Feckler, A., Graf, T., Joschko, T. J., Jupke, J., Noss, C., Rösch, V., Schirmel, J., Schmidt, T., Zubrod, J. P., & Schulz, R. (2022). Structural and functional development of twelve newly established floodplain pond mesocosms. *Ecology and Evolution*, 12(3), e8674. <https://doi.org/10.1002/ece3.8674>